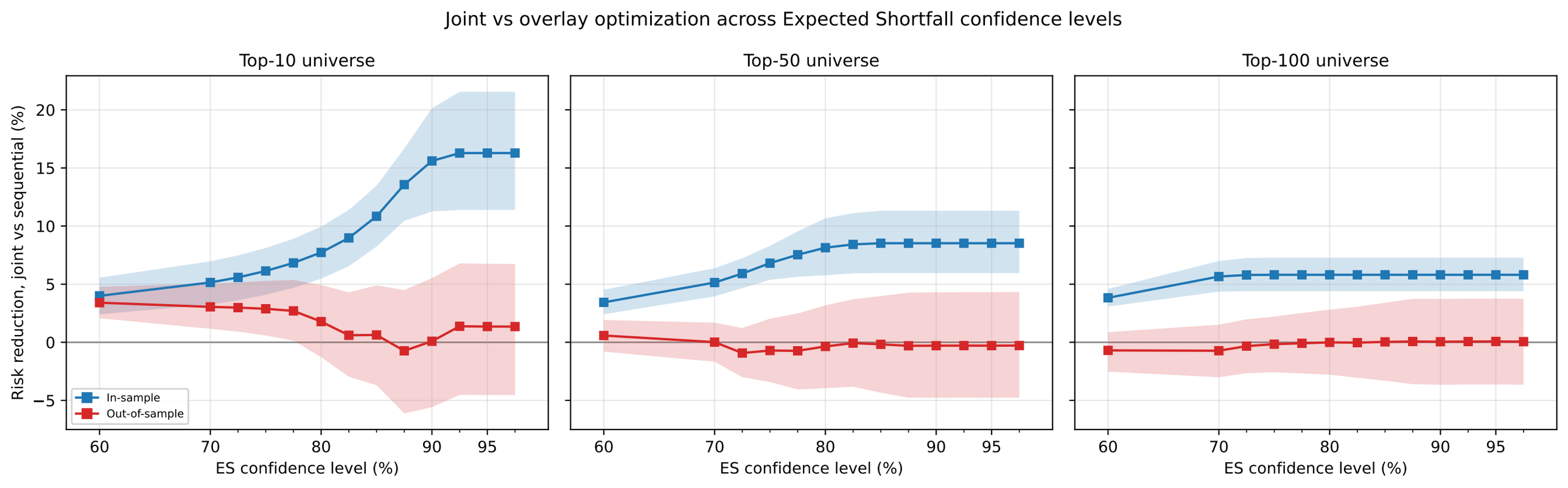
Joint vs overlay optimization across Expected Shortfall confidence levels
Top-10 universe
Top-50 universe
Top-100 universe
Risk reduction, joint vs sequential (%)
20
15
10
5
0
−5
In-sample
Out-of-sample
60
70
80
90
95
ES confidence level (%)